KERALA POLYTECHNIC STUDY HUB

Applied Physics-I

Subject Code: **1003** | Diploma in Engineering and Technology | Semester 1 | Credits: 3

Student-friendly notes with English + Malayalam explanations, formulas, diagrams, solved problems, answers, hints and quick revision support.

Expected questions are practice-focused, not a guarantee of actual exam questions.

Subject
Applied Physics-I

Code
1003

Credits
3

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Course Overview

Course Code 1003 | Semester 1 | Basic Science

How to study

1. Read the English idea first, then the Malayalam support line.
2. Write formula, symbol meaning and SI unit together: example, **F = ma**, SI unit **N**.
3. Practise numericals using Given, Formula, Substitution and Answer.
4. Draw diagrams for vectors, circular motion, heat transfer, capillary rise and Bernoulli flow.

Course roadmap

1. Measurements, vectors, equations of motion and momentum.
2. Circular motion, banking, moment of inertia and angular momentum.
3. Work, energy, power, friction, heat and temperature.
4. Elasticity, pressure, surface tension, viscosity and fluid motion.

Physics map: Units → Vectors → Motion → Force → Rotation → Energy → Heat → Fluids

MINI FORMULA BANK

Important formulas with proper symbols

Topic	Formula	Unit / note
Error	Absolute error = |measured − true|; relative error = absolute error / true value; % error = relative error × 100	relative error has no unit; percentage error is %
Vectors	$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$; $A_x = A\cos\theta$; $A_y = A\sin\theta$	same unit as vector
Motion	$v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$	s: m, u/v: m s^{-1}, a: m s^{-2}
Force and momentum	$F = ma$; $p = mv$; $J = F\Delta t = \Delta p$	N, kg m s^{-1}, N s
Circular motion	$\omega = \theta/t$; $\alpha = \Delta\omega/\Delta t$; $v = r\omega$; $a_c = v^2/r = r\omega^2$; $F_c = mv^2/r$	rad s^{-1}, rad s^{-2}, m s^{-1}, N
Banking	$\tan\theta = v^2/rg$; $v = \sqrt{rg\tan\theta}$	θ in degree/radian; v in m s^{-1}
Moment of inertia	$I = \Sigma mr^2$; $I = Mk^2$; $\tau = rF\sin\theta$; $L = I\omega$	kg m^2, N m, kg m^2 s^{-1}
Work and energy	$W = Fs\cos\theta$; $KE = \frac{1}{2}mv^2$; $PE = mgh$; $P = W/t = Fv$	J, J, J, W
Heat	$K = °C + 273$; $°F = (9/5)°C + 32$; $Q = mc\Delta T$	K, °F, J
Elasticity	Stress = F/A; strain = ΔL/L; Y = stress/strain	Pa, no unit, Pa
Fluids	$P = F/A$; $h = 2T\cos\theta/(\rho gr)$; $F = 6\pi\eta rv$; $Re = \rho vD/\eta$; $A_1v_1 = A_2v_2$	Pa, m, N, no unit, m^3 s^{-1}

Bernoulli	$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$	each term: Pa or J m^{-3}

MODULE 1

Measurements, Vectors and Motion

Malayalam idea: അളവുകൾ, units, errors, vectors, motion, force, momentum എന്നിവയാണ് ഈ module-ന്റെ അടിസ്ഥാനം. Formula എഴുതുമ്പോൾ SI unit നിർബന്ധമായി ചേർക്കുക.

Physical quantities and units

A physical quantity is any measurable property. Fundamental quantities are length, mass, time, current, temperature, amount of substance and luminous intensity. Derived quantities are obtained from these, such as area, velocity, force and pressure.

Malayalam: അളക്കാൻ കഴിയുന്ന ഗുണമാണ് physical quantity. Length, mass, time പോലുള്ളവ fundamental; velocity, force പോലുള്ളവ derived quantities ആണ്.

```
1 N = 1 kg m s⁻²
1 J = 1 N m
1 Pa = 1 N m⁻²
```

Application: workshop measurement, surveying, electrical readings and machine design.

Errors in measurement

No measurement is perfectly exact. Absolute error is the difference between measured and true value. Relative error is absolute error divided by true value.

Malayalam: Measured value-നും true value-നും ഉള്ള വ്യത്യാസമാണ് absolute error.

$$\text{Absolute error} = |x_m - x_{true}|$$

$$\text{Relative error} = \text{absolute error} / \text{true value}$$

$$\text{Percentage error} = \text{relative error} \times 100$$

Scalars, vectors and resolution

Scalars have magnitude only; vectors have magnitude and direction. Resolution splits a vector into perpendicular components.

Malayalam: Magnitude മാത്രം ഉള്ളത് scalar. Magnitude കൂടെ direction ഉള്ളത് vector.

$$A_x = A \cos \theta$$

$$A_y = A \sin \theta$$

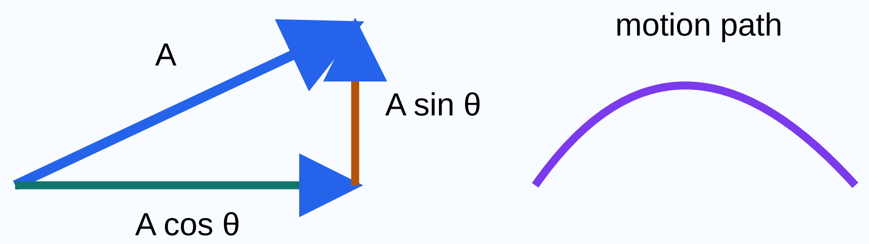
$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

Motion and Newton laws

Newton's first law explains inertia, second law gives F = ma, and third law gives action-reaction. Momentum is mass times velocity.

Malayalam: Force = mass × acceleration. External force ഇല്ലെങ്കിൽ total momentum constant ആണ്.

```
v = u + at
s = ut + ½at²
v² = u² + 2as
p = mv; J = FΔt = Δp
```



Solved problems

1. A rod is measured 2.04 m instead of 2.00 m.

   Answer: Absolute error = 0.04 m; relative error = 0.04/2 = 0.02; percentage error = 2%.
2. A body starts from rest with a = 2 m s^{-2} for 5 s.

   Answer: v = 0 + 2 × 5 = 10 m s^{-1}. s = ½ × 2 × 25 = 25 m.
3. A 5 kg body moving at 4 m s^{-1} stops in 0.2 s.

   Answer: |F| = Δp/Δt = 20/0.2 = 100 N.

Try yourself and expected questions

- Resolve 100 N at $\theta = 30°$.

  Answer: $A_x = 100 \cos 30° = 86.6$ N; $A_y = 100 \sin 30° = 50$ N.
- State triangle law and parallelogram law of vector addition.

  Hint: Draw head-to-tail triangle and parallelogram diagonal; resultant is the closing side/diagonal.
- Explain recoil of gun and rocket propulsion using conservation of momentum.

  Hint: Write total momentum before = total momentum after; bullet/gas moves forward/backward, gun/rocket moves opposite.
- Long answer: Newton's laws with examples, formulas and applications.

  Hint: Use three headings: inertia, F = ma, action-reaction; add seat belt, pushing trolley and rocket examples.

MODULE 2

Circular Motion and Rotational Dynamics

Malayalam idea: Circular motion-ൽ direction മാറുന്നതിനാൽ centre-ലേക്ക് acceleration ഉണ്ടാകും. Rotational motion-ൽ mass distribution വളരെ പ്രധാനമാണ്.

Angular quantities

Angular displacement is angle swept by radius. Angular velocity is rate of angular displacement. Angular acceleration is rate of change of angular velocity.

Malayalam: Radius vector sweep ചെയ്യുന്ന angle ആണ് angular displacement.

```
ω = θ/t
α = Δω/Δt
v = rω; a = rα
```

Centripetal force and banking

Centripetal acceleration and force act towards the centre. Banking raises the outer edge of road/rail to provide safe centripetal force.

Malayalam: Centre-ലേക്ക് acting ചെയ്യുന്ന force ആണ് centripetal force. Banking skidding കുറയ്ക്കുന്നു.

$$a_c = v^2/r = r\omega^2$$

$$F_c = mv^2/r = mr\omega^2$$

$$\tan\theta = v^2/rg$$

Moment of inertia

Moment of inertia is rotational inertia. Radius of gyration is the distance at which whole mass may be assumed concentrated.

Malayalam: Rotational motion മാറ്റാൻ ഉള്ള resistance ആണ് moment of inertia.

```
I = Σmr²
I = Mk²
k = √(I/M)
```

Body	Moment of inertia
Rod about centre	$ML^2/12$
Rod about end	$ML^2/3$
Ring	MR^2
Disc	$MR^2/2$
Solid sphere	$2MR^2/5$
Hollow sphere	$2MR^2/3$

Axis theorems, torque and angular momentum

Parallel axis theorem: $I = I_G + Md^2$. Perpendicular axis theorem: $I_z = I_x + I_y$. Torque is turning effect of force. Angular momentum is conserved when external torque is zero.

Malayalam: External torque ഇല്ലെങ്കിൽ angular momentum constant ആയിരിക്കും.

```
τ = rF sin θ
L = Iω
If τext = 0, then Iω = constant
```

Solved problems

1. Wheel radius r = 0.5 m, ω = 20 rad s^{-1}.
   Answer: $v = r\omega = 0.5 \times 20 = 10$ m s^{-1}.
2. Mass m = 2 kg, r = 1 m, v = 4 m s^{-1}.
   Answer: $F_c = mv^2/r = 2 \times 16/1 = 32$ N.
3. Disc mass M = 4 kg, radius R = 0.2 m.
   Answer: $I = MR^2/2 = 4 \times 0.04/2 = 0.08$ kg m^2.

Practice

- Find τ for F = 50 N at perpendicular distance r = 0.4 m.

  Answer: $\tau = rF = 0.4 \times 50 = 20$ N m.
- State parallel and perpendicular axis theorems.

  Answer: Parallel: $I = I_G + Md^2$. Perpendicular: $I_z = I_x + I_y$.
- Explain physical significance of moment of inertia.

  Hint: Compare with mass in linear motion; larger I means harder to change rotational speed.
- Long answer: banking of roads with formula and diagram.

  Hint: Draw raised outer edge, write $\tan \theta = v^2/rg$, mention safe turning and reduced skidding.

MODULE 3

Work, Energy, Power, Heat and Temperature

Malayalam idea: Force displacement ഉണ്ടാക്കുമ്പോൾ work done. Energy work ചെയ്യാനുള്ള capacity. Heat temperature difference കാരണം transfer ചെയ്യുന്ന energy ആണ്.

Work, friction and power

Work is done when force produces displacement. Friction opposes relative motion. Static friction acts before motion; kinetic friction acts during motion. Power is rate of doing work.

Malayalam: Displacement ഇല്ലെങ്കിൽ work zero. Friction motion oppose ചെയ്യുന്നു.

```
W = Fs cos θ
f = μR
P = W/t = Fv
```

Reducing friction: lubrication, polishing, ball bearings, streamlining.

Energy

Kinetic energy is due to motion; potential energy is due to position. In free fall, potential energy changes into kinetic energy, while total mechanical energy remains constant.

Malayalam: Falling body-ൽ PE കുറയുകയും KE കൂടുകയും ചെയ്യും.

```
KE = ½mv²
PE = mgh
Total energy = KE + PE = constant
```

Heat and temperature

Heat is energy transferred due to temperature difference. Temperature measures hotness or coldness. Heat transfer occurs by conduction, convection and radiation.

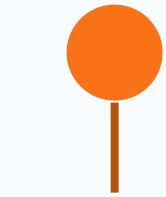
Malayalam: Conduction solids-ൽ, convection fluids-ൽ, radiation medium ഇല്ലാതെയും നടക്കും.

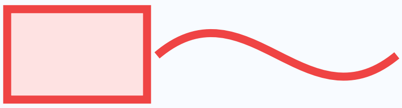
```
K = °C + 273
°F = (9/5)°C + 32
Q = mcΔT
```

Specific heat and thermometers

Specific heat is heat required to raise the temperature of unit mass by 1 K. Mercury thermometer uses expansion of mercury. Pyrometer measures high temperature without contact.

Malayalam: Furnace temperature പോലുള്ള high temperature pyrometer ഉപയോഗിച്ച് measure ചെയ്യാം.





Solved problems

1. Force $F = 20$ N moves body $s = 5$ m.

   Answer: $W = Fs = 20 \times 5 = 100$ J.
2. Mass $m = 2$ kg, velocity $v = 10$ m s^{-1}.

   Answer: $KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 2 \times 100 = 100$ J.
3. Convert 40 °C.

   Answer: °F = (9/5) × 40 + 32 = 104 °F; K = 40 + 273 = 313 K.

Practice

- Find power if 600 J work is done in 20 s.

  Answer: $P = W/t = 600/20 = 30$ W.
- Find PE of 5 kg body at $h = 3$ m.

  Answer: $PE = mgh = 5 \times 9.8 \times 3 = 147$ J.
- Explain conduction, convection and radiation with examples.

  Hint: conduction: hot spoon; convection: boiling water; radiation: heat from Sun.
- Long answer: conservation of energy for freely falling body.

  Hint: Show $PE = mgh$ at top, $KE = \frac{1}{2}mv^2$ during fall, and PE + KE remains constant if air resistance is neglected.

MODULE 4

Elasticity, Pressure, Surface Tension and Fluids

Malayalam idea: Elasticity material deformation explain ചെയ്യുന്നു. Fluids-ൽ pressure, surface tension, viscosity, streamline flow, Bernoulli theorem എന്നിവ പ്രധാനമാണ്.

Elasticity

Elasticity is the property of a body to regain original shape after removing deforming force. Stress is force per unit area. Strain is change/original dimension. Hooke's law: stress is proportional to strain within elastic limit.

Malayalam: Elastic limit ഉള്ളിൽ stress strain-ന് directly proportional ആണ്.

```
Stress = F/A
Strain = ΔL/L
Y = stress/strain
```

Pressure

Pressure is normal force per unit area. Atmospheric pressure is due to air column. Gauge pressure is above atmospheric pressure. Absolute pressure is measured from vacuum.

Malayalam: Absolute pressure = atmospheric pressure + gauge pressure.

$$P = F/A$$

$$P_{absolute} = P_{atm} + P_{gauge}$$

Surface tension

Surface tension makes the liquid surface behave like a stretched membrane. Cohesive force acts between same molecules; adhesive force between different molecules. Capillary rise occurs due to surface tension.

Malayalam: Liquid surface stretched membrane പോലെ behave ചെയ്യുന്നതാണ് surface tension.

```
T = F/l
h = 2T cos θ/(ρgr)
```

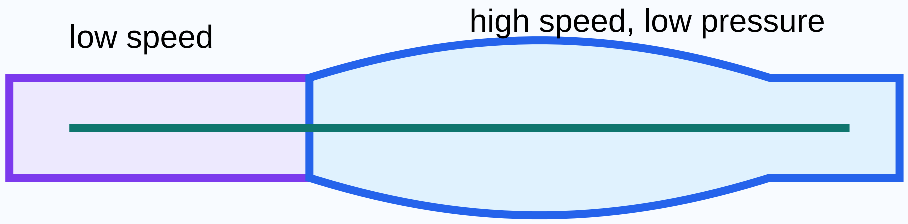
Temperature usually decreases surface tension; detergents also reduce it.

Viscosity and fluid flow

Viscosity is resistance to relative motion between fluid layers. Stokes law gives viscous force. Reynolds number predicts streamline or turbulent flow. Continuity equation states $A_1v_1 = A_2v_2$.

Malayalam: Narrow pipe section-ൽ velocity കൂടും. High velocity region-ൽ pressure കുറയുന്നത് Bernoulli theorem explain ചെയ്യുന്നു.

```
F = 6πηrv
Re = ρvD/η
A₁v₁ = A₂v₂
P + ½ρv² + ρgh = constant
```



Solved problems

1. 100 N force on 0.02 m^2 area.

   Answer: Stress = F/A = 100/0.02 = 5000 Pa.
2. Wire length 2 m extends by 1 mm.

   Answer: Strain = $\Delta L/L$ = 0.001/2 = 0.0005.
3. Pipe area A_1 = 4 cm^2 has speed v_1 = 2 m s^{-1}; area A_2 = 2 cm^2.

   Answer: $v_2 = A_1v_1/A_2$ = 4 × 2/2 = 4 m s^{-1}.

Practice

- Find pressure for F = 500 N on A = 0.25 m^2.

  Answer: P = F/A = 500/0.25 = 2000 Pa.
- State Hooke's law and define Young's modulus.

  Answer: Hooke's law: stress ∝ strain within elastic limit. Y = stress/strain.
- Explain applications of surface tension.

  Hint: mention capillary tubes, ink flow, detergents, paints, soldering and lamp wick.
- Long answer: Bernoulli theorem with airfoil and atomiser applications.

  Hint: Write $P + \frac{1}{2}\rho v^2 + \rho gh$ = constant; faster fluid means lower pressure; use this to explain lift/spray.

QUICK REVISION

Last-minute Revision Sheet

Memory tricks

- **v-u-at:** final velocity = initial velocity + acceleration effect.
- **F-ma:** more mass needs more force for same acceleration.
- **v-rω:** rim speed increases with radius.
- **PE → KE:** falling body converts stored energy into motion energy.
- **Av constant:** narrow pipe gives faster flow.

Common mistakes

- Writing answer without SI unit.
- Not converting cm to m or minutes to seconds.
- Adding vectors like scalars.
- Confusing mass and weight.
- Using gauge pressure where absolute pressure is needed.

Expected exam-style questions with hints/answers

1. Define fundamental and derived quantities with examples.

   Answer: fundamental quantities are independent basic quantities like length, mass, time; derived quantities are formed from them, like velocity and force.
2. State Newton's laws and explain impulse.

   Hint: write inertia, $F = ma$, action-reaction; impulse $J = F\Delta t = \Delta p$.
3. Explain recoil of gun and rocket propulsion.

   Hint: use conservation of momentum; forward bullet/gas gives backward/forward reaction motion.
4. Derive or state centripetal force and explain banking of roads.

   Answer: $F_c = mv^2/r$; banking supplies centripetal force and reduces skidding.
5. State parallel and perpendicular axis theorems.

   Answer: $I = I_G + Md^2$ and $I_z = I_x + I_y$.
6. Explain work, energy, power and conservation of energy.

   Hint: include $W = Fs \cos \theta$, $KE = \frac{1}{2}mv^2$, $PE = mgh$, $P = W/t$ and $PE + KE =$ constant.
7. Explain conduction, convection and radiation.

   Answer: conduction needs material contact, convection happens by fluid circulation, radiation travels without medium.
8. State Hooke's law and define elastic moduli.

   Answer: stress $\propto$ strain within elastic limit; Y, K and rigidity modulus compare stress with corresponding strain.
9. Explain surface tension, capillary rise and applications.

   Hint: define $T = F/l$, write $h = 2T \cos \theta/(\rho g r)$, add wick, ink, soil water and detergents.
10. State Bernoulli's theorem and explain airfoil or atomiser.

    Answer: $P + \frac{1}{2}\rho v^2 + \rho g h =$ constant; higher speed gives lower pressure, causing lift or spray action.

Long answer writing practice

Hint: Use this order: definition, labelled diagram, explanation, formula, symbol meaning, SI unit, real-life example, engineering application and common mistake.

Applied Physics-I 1003 - Simple HTML Notes

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